Math 4141: Geometry and Differential Calculus

Dr. Md Tusher Mollah

Assistant Professor, Department of Natural Sciences
Islamic University of Technology, Bangladesh
PhD and Postdoc from the Department of Civil and Mechanical Engineering

Math 4141: Geometry and Differential Calculus **Credit: 4.00**

Course Contents:

2D Coordinate Geometry: Change of axes: Transform of coordinates, Simplification of equations of curves; Pair of straight lines: Homogeneous second-degree equations, Conditions for general second-degree equations to represent a pair of straight lines, Angle between the lines, Pair of straight lines joining the origin to the points of intersection of the curve and a line, Circles and system of circles: Tangent and normal, Radial axis and its properties, Parametric coordinates.

3D Coordinate Geometry: Rectangular coordinates, Direction cosines and angle between two lines, The plane and the straight lines, The equation of a sphere, The standard forms of equations of the central conicoid, cones and cylinders.

Differential Calculus: Limits, Continuity and Differentiability; Differentiation of explicit and implicit functions and parametric equations; Significance of derivatives, Differentials; Successive differentiation of various types of functions; Leibnitz's Theorem; Rolle's Theorem; Mean Value Theorem; Taylor's Theorem in finite and infinite forms; Maclaurin's Theorem in finite and infinite forms; Lagrange's form of remainders; Cauchy's form of remainders; Expansion of functions by differentiation and integration; Partial differentiation; Euler's Theorem; Tangent, maximum and minimum values of functions and points of inflection; Applications of Differential Calculus; Evaluation of indeterminate forms by L'Hospitals rule; Curvature, the center of curvature and chord of curvature; Evolutes and involutes; Asymptotes, Envelopes, Curve tracing.

Pair of straight Lines

Pair of straight Lines: $ax^2 + 2hxy + by^2 = 0$

Angle between the line: $\tan\theta = \frac{2\sqrt{h^2-ab}}{a+b}$

If pair of straight lines are perpendicular: $a + b = 0$

If pair of straight lines are parallel (coincide): $h^2 = ab$

The equation of a **general pair of straight lines** is the product of the equations of the two lines:

$(A_1x + B_1y + C_1)(A_2x + B_2y + C_2) = 0$. Each factor represents one line. Expanding gives a general second-degree equation:

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, with coefficients that make it factor into those two lines.

If both lines pass through the origin, then $C_1 = C_2 = 0$, so the equation is homogeneous:

$(A_1x + B_1y)(A_2x + B_2y) = 0$. For example, $(x + 3y - 1)(x + 3y + 5) = 0$ is a pair of distinct parallel lines.

General Equation of Second degree (Class work)

The general equation of second degree $\boldsymbol{ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0}$ will represent

1 (i) a pair of straight lines if $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$

(ii) Two parallel lines if $\Delta = 0, ab = h^2$ (no finite point of intersection)

(iii) Two perpendicular lines if $\Delta = 0, a + b = 0$ (From the angle between two curves)

2. a circle if $a = b \neq 0, h = 0$

3. a parabola if $ab = h^2, \Delta \neq 0$

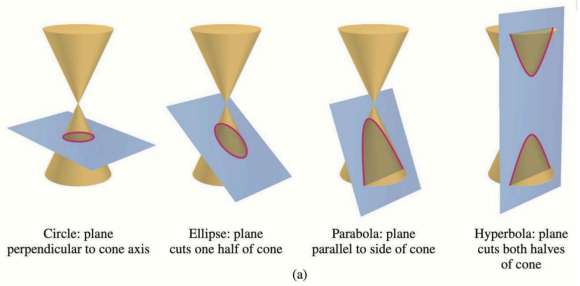
4. an ellipse if $ab - h^2 > 0, \Delta \neq 0$

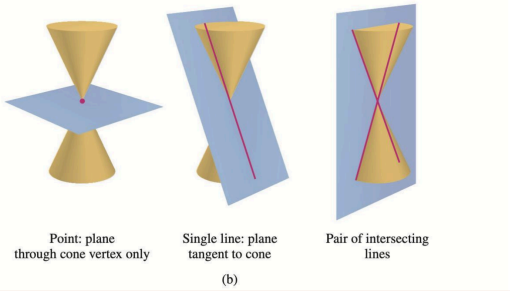
5. a hyperbola if $ab - h^2 < 0, \Delta \neq 0$.

Graphical Representation

A **cone** is a three-dimensional surface. For example, $x^2 + y^2 = z^2$ is a double cone: it has an upper and a lower part meeting at the vertex (0,0,0).

A **conic** is a two-dimensional curve. It is formed when a plane cuts through a cone.





Exercise

What does the equation $5x^2$-2xy+$5y^2$-8x-8y-8=0 represent?

Solution:

The given equation $5x^2$-2xy+$5y^2$-8x-8y-8=0 ……(1)

Comparing (1) with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

We get a=5, h=-1,b=5,g=-4,f=-4,c=-8

$$\therefore \Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$$

$= abc + 2fgh - af^2$-bg^2-ch^2

=-200-32-80-80+8

=-384

$\therefore \Delta \neq 0$

$ab - h^2 = 5.5 - (-1)^2$=25-1=24

$ab - h^2 > 0$.

Thus the given equation represents an ellipse.

(HW) Prove that the equation $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$ represent a pair of parallel lines. Find equation of straight lines.

Solution: Given $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$

Now compare with equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ then we get

$a = 1, h = 3, b = 9, g = 2, f = 6$ and $c = -5$

$\Delta = abc + 2fgh - af^2$-$bg^2$-$ch^2 = 0$

We have $\Delta = 0$

$$\tan\theta = \frac{2\sqrt{h^2 - ab}}{a + b} \Rightarrow \theta = 0$$

Now $x^2 + 6xy + 9y^2 + 4x + 12y - 5 = 0$

$$\Rightarrow x^2 + x(6y + 4) + 9y^2 + 12y - 5 = 0$$

$$\Rightarrow x = \frac{-(6y+4) \pm \sqrt{(6y+4)^2 - 4(9y^2 + 12y - 5)}}{2.1}$$

$\therefore x + 3y + 5 = 0$ and $x + 3y - 1 = 0$